

FOUNDATIONS OF GROMOV–WITTEN THEORY

1. LIST OF TALKS

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2. DESCRIPTION OF TALKS

1. **Riemann surfaces and Hurwitz numbers.** Pages 1–12 of the notes [1] may serve as a rough guideline for the talk. The same material is treated much more detailed in Chap 1–9 in the book [2]. The following aspects should be covered in the talk since they will become relevant later:
 - What is a Riemann surface (or a smooth algebraic curve over the complex numbers if you prefer algebraic geometry)?
 - What is the genus of a Riemann surface?
 - What is a morphism of Riemann surfaces?
 - How are Hurwitz numbers defined and what do they count?
 - State Burnside’s formula for Hurwitz numbers.

2. **Intersection theory.** The relevant material can be found in Chapter 9 and 10 of lecture notes of Gathmann [3]. Also Chapter 1 and 9 of Eisenbud and Harris' book [19] might be helpful to you. You do not need to cover all material presented in the first reference but the following points should be covered in any event since they will become relevant later in the seminar:

- The definition of Chow groups
- The Chow group of projective space
- The existence of proper push-forwards and flat pull-backs
- The existence of an intersection product [19, Theorem–Definition 1.5]
- Intuition on Chern classes, most importantly the feature described in [3, Proposition 10.2.6] and its generalisation in [3, Proposition 10.3.12]
- The calculation that there are 27 lines on a cubic surface in $\mathbb{P}^3$ [3, Example 10.3.15]

3. **Intersection theory on the moduli space of curves.** Introduce the moduli space of pointed curves $\mathcal{M}_{g,n}$ and the Deligne–Mumford compactification $\overline{\mathcal{M}}_{g,n}$. In particular, illustrate $\overline{\mathcal{M}}_{0,n}$ and $\overline{\mathcal{M}}_{1,1}$ [4, Sec. 1]. Discuss forgetful and attaching maps briefly and explain how $\overline{\mathcal{M}}_{g,n+1}$ can be identified with the universal curve over $\overline{\mathcal{M}}_{g,n}$ [4, Sec. 2.1]. Introduce ψ and λ -classes [4, Sec. 2.2 & 2.3.3] and compute some basic intersection numbers:

$$\int_{\overline{\mathcal{M}}_{0,n}} \psi_1^{k_1} \cdots \psi_n^{k_n}, \quad \int_{\overline{\mathcal{M}}_{1,1}} \psi_1, \quad \int_{\overline{\mathcal{M}}_{1,1}} \lambda_1.$$

The first follows from the so-called string equation (cf. [20, Lem. 1.4] or [6, p. 514]).

4. **Moduli space of genus zero stable maps.** Introduce the moduli space $\overline{\mathcal{M}}_{0,n}(X, \beta)$ of genus-zero stable maps to a smooth projective variety X in curve class $\beta \in H_2(X, \mathbb{Z})$. Focus on the case $X = \mathbb{P}^2$ and discuss the illustrating example $\beta = 2H$ where H is the class of a line. Discuss the boundary of $\overline{\mathcal{M}}_{0,n}(X, \beta)$. Prove Kontsevich's recursion for the number of degree- d rational curves through $3d - 1$ points in $\mathbb{P}^2$. When X is a general cubic surface on $\mathbb{P}^3$, reduce $\int_{[\overline{\mathcal{M}}_{0,0}(X, H)]} 1$ to the calculation done at the end of talk 2 (cf. [3, Example 10.3.15]). Apart from the last point, these notes [5] may serve as a rough guideline for your talk. All material can be found in more detail in [21, Chap. 2–3] and [22, Sec. 0, 1, 6, 9].

5. **Moduli space of higher genus stable maps.** Introduce the moduli space $\overline{\mathcal{M}}_{g,n}(X, \beta)$ of genus- g stable maps to a smooth projective variety X [6, Chap. 24]. Explain that there are several issues with this moduli space by considering the following concrete examples:

- $\overline{\mathcal{M}}_{1,0}(\mathbb{P}^1, [\mathbb{P}^1])$ [9]
- $\overline{\mathcal{M}}_{1,0}(\mathbb{P}^2, H)$ [9]
- $\overline{\mathcal{M}}_{1,0}(\mathbb{P}^2, 3H)$ [7, Ex. 8.4]

Use this to motivate the formation of a virtual fundamental class. Heuristically calculate the expected dimension of this class. Explain the mismatch between $[\overline{\mathcal{M}}_g(X, 0)]^{\text{vir}}$ and $[\overline{\mathcal{M}}_g] \times [X]$ [7, Ex. 8.3]. Motivate the formula for the virtual fundamental class when the moduli space is smooth (and some

other technical assumptions are met) [8, Lem. 6.1.1]. Use this to calculate $[\overline{\mathcal{M}}_{1,0}(\mathbb{P}^2, H)]^{\text{vir}}$ [9].

6. **Equivariant intersection theory and localisation.** Introduce equivariant cohomology groups $H_G^*(X)$ of a variety with the action of a group G following for instance [10, Chap. 7]. For our purposes it is enough to consider $G = (\mathbb{C}^\times)^n$. Compute

$$(1) \quad H_{(\mathbb{C}^\times)^n}^*(\text{pt}) \quad \text{and} \quad H_{(\mathbb{C}^\times)^n}^*(\mathbb{P}^{n-1}).$$

Explain that essentially all features about usual intersection theory covered in talk 2 lift to equivariant cohomology. (One technical difference is that talk 2 discussed Chow groups. See [11] for a construction of equivariant Chow groups.) State the Atiyah–Bott localisation formula [10, Thm. 8.5.1]. Later we will crucially the formula require it in Gromov–Witten calculations. Illustrate the strength of the formula by showcasing two calculations [10, Lem. 8.5.7 & Ex. 8.5.12]. If time permits you can present aspects of the proof of the formula. See [12, Sec. 3] for an alternative treatment of these topics.

7. **Localisation for stable maps to projective space.** The goal of this talk is to apply the Atiyah–Bott localisation formula from the last talk (or actually its upgrade to virtual fundamental classes [13]) to the moduli space of stable maps to $\mathbb{P}^n$. This is discussed in [13, Sec. 4] with more details spelled out in [6, Chap. 27] and [12, Sec. 4]. The specialisation of this formula to $n = 1$ [16, Prop. 1] will be crucial in the next talk and should hence be mentioned explicitly. Using the formula proven for ψ -class integrals in talk 3, argue that virtual localisation completely solves GW theory of projective space in genus zero. Redo the calculation of the 27 lines on a cubic surface via Atiyah–Bott localisation [9].
8. **The ELSV formula and relative stable maps.** The main player of this talk is the Ekedahl–Lando–Shapiro–Vainshtein formula relating Hurwitz numbers to linear hodge integrals [14, 2.2. Thm]. You probably need to recall some aspects about Hurwitz numbers from the first talk. (See the references there.) For the proof of the formula you can follow [14, Sec. 5]. The proof idea presented there relies on the existence of a moduli space of so-called relative stable maps to $\mathbb{P}^1$ which was constructed in algebraic geometry just after the paper appeared. This moduli space and the resulting proof of the ELSV formula is presented in much detail in [15] and summarised briefly in [1, Sec. 6]. Relative stable maps will play a prominent role in talk 11 as well. So please include a brief discussion of these in the talk.
9. **Hurwitz numbers, Hodge integrals and Fock space.** Introduce infinite wedge space. Everything we may require is summarised in [16, Sec. 2.0]. See [17, Sec. 5–7]. Following [16, Sec. 2.1], express linear Hurwitz numbers enumerating branched covers of $\mathbb{P}^1$ with prescribed ramification μ over one point and simple ramification over b points as a vacuum expectation value on Fock space. For this you need to recall Burnside’s formula (cf. talk 1).

Via the ELSV formula, express linear Hodge integrals as vacuum expectation values. In particular, arrive at the formula in [16, Eq. (2.17)]. If time permits you can illustrate its strength by showcasing the calculation [16, Prop. 10].

10. **The stationary Gromov-Witten theory of $\mathbb{P}^1$.** The goal is to prove (or at least to motivate the proof of) [16, Prop. 11] which expresses the (absolute stationary nonequivariant) Gromov–Witten invariants of $\mathbb{P}^1$ as a vacuum expectation value on Fock space. The proof greatly relies on talk 7 and 9. I will provide a more detailed guideline closer to the talk. If time permits, aspects of [16, Sec. 4] can be discussed as well but they will not be relevant for the last talk.
11. **The Gromov-Witten/Hurwitz correspondence.** The goal is to prove the correspondence of the title which relates stationary relative Gromov–Witten invariants of a smooth curve X with Hurwitz numbers enumerating branched covers of X . It is stated in [18, Thm. 1] and its proof is the content of [18, Sec. 1]. I will provide a more detailed guideline closer to the talk.

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